AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2016

Chemistry Paper II

Time: 2 hours 25 minutes Marks: 40

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

a. $C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(l)}$ (2 Marks)

How many grams of oxygen gas (O_2) will be required to react completely with 3 moles of propane gas (C_3H_8)?

(Note: Atomic mass of carbon = 12 amu, hydrogen = 1 amu and oxygen = 16 amu)

b. Balance the given chemical equations. (2 Marks)

i. $___C_2H_{6(g)} + ___O_{2(g)} \rightarrow ___CO_{2(g)} + ___H_2O_{(l)}$

ii. $___Cu_2S_{(s)} + ___O_{2(g)} \rightarrow ___Cu_2O_{(s)} + ___SO_{2(g)}$

Q.2. (Total 3 Marks)

How many protons, neutrons and electrons are there in the ion $^{31}_{15}P^{3-}$?

Ion	Number of Protons	Number of Electrons	Number of Neutrons
$^{31}_{15}P^{3-}$			

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Q.3. (Total 5 Marks)

Following is a sketch of the modern periodic table showing a few representative elements. With reference to each element, answer the given questions.

Group →	**I**	**II**	**III**	**IV**	**V**	**VI**	**VII**	**VIII**
Period ↓								
1								He
2	Li	Be		C		O		
3			Al		P			

a. Identify the number of shells in an aluminium atom. (1 Mark)

b. Why is beryllium less reactive than lithium? (1 Mark)

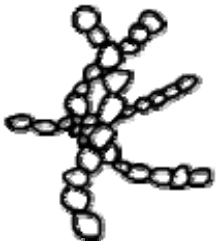
c. What happens to shielding effect of elements if we move across period 2 from left to right? (1 Mark)

d. Amongst the given elements, oxygen is the most electronegative. Why is helium not the most electronegative element? (1 Mark)

e. Which element has the lowest first ionisation energy? (1 Mark)

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Q.4. (Total 4 Marks)

	
A	B

Identify the given allotropic forms of sulphur. Give ONE property of each.

A:_____

Property: _____

_____

_____

B: _____

Property: _____

_____

_____

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Q.5. (Total 5 Marks)

a. Draw a labelled diagram of an electrolytic cell to show the electroplating of an iron nail with zinc. Your diagram should include: (3 Marks)

i. Name of the electrolyte

ii. Chemical equation to show the chemical change that occurs at anode

b. Why is iron nail used as cathode? (1 Mark)

_____

_____

c. What is the benefit of electroplating iron nail with zinc? (1 Mark)

_____

_____

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Q.6. (Total 4 Marks)

Identify the given elements as either an alkali metal or an alkaline earth metal.

i. Beryllium: _____

ii. Rubidium: _____

iii. Radium: _____

iv. Barium: _____

Q.7. (Total 8 Marks)

EITHER

a.

i. What are isotopes?

ii. Name three types of isotopes of carbon.

iii. Draw the atomic structures of three isotopes of carbon specifying their number of protons, neutrons and electrons.

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OR

b.

i. Draw dot and cross structures of the given species and identify the type of bonding present in each.

ii. Write any FOUR characteristics of ionic compounds.

Species	Dot and Cross Structure	Type of Bond
NH_4^+		
Li_2O		

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Q.8. (Total 7 Marks)

EITHER

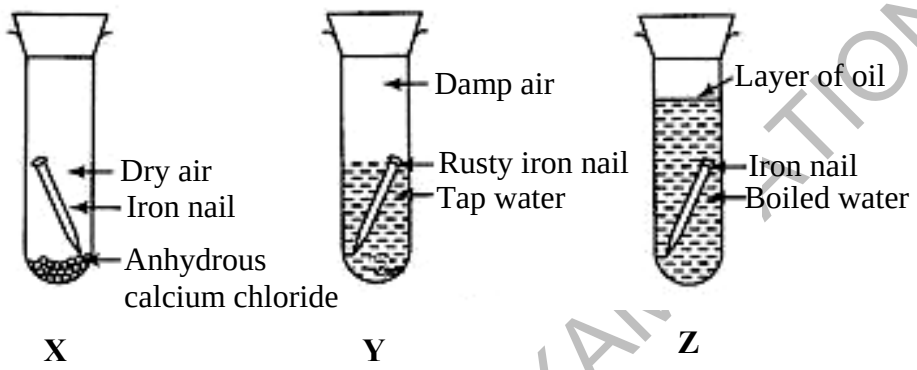
a. A student takes equal volume of water in three beakers **X**, **Y** and **Z**. She then mixes talcum powder in beaker **X**, egg albumen in beaker **Y** and ethanol in beaker **Z**.

i. Using the terms solution, suspension and colloid, identify the type of solution formed in each beaker **X**, **Y** and **Z**. Each term can be used once, twice or not at all.

ii. Write any FOUR distinguishing characteristics of solutions **Y** and **Z**.

OR

b. Identical iron nails are kept in three test tubes **X, Y** and **Z** as shown below.



X **Y** **Z**

i. Why does the iron nail in test tube **Y** becomes rusty but the ones in test tubes **X** and **Z** remain unaffected? Explain with reference to each test tube.

ii. Describe any TWO methods used in everyday life to prevent iron objects from rusting.

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